

12.0 Summative Assessment: Cabling and Physical Network Infrastructure

For this summative assessment, I selected Packet Tracer Lab 12.10 – Implement Basic Network Connectivity because it provided the most comprehensive hands-on experience within the competency. This lab required configuring network devices, verifying connectivity, troubleshooting configuration issues, and applying networking concepts learned throughout previous labs without relying entirely on guided instructions. The activity reinforced foundational networking skills involving physical installation considerations, device configuration, interface verification, command-line management, and troubleshooting connectivity problems between network devices.

One of the most important concepts reinforced during this lab was the role physical installation plays in overall network reliability and performance.

Important physical installation factors included:

- Proper cable selection for device connectivity
- Interface verification and operational status
- Cable management and organization
- Stable power and hardware readiness
- Accurate physical device connections

Correct cable types must be selected based on the devices being connected. Network administrators must understand the differences between straight-through, crossover, console, and fiber connections to ensure successful communication between devices. Incorrect cable selection can create Layer 1 connectivity failures that prevent devices from communicating properly.

The lab also emphasized the importance of verifying physical interfaces and ensuring ports are active and operational. Interfaces that are administratively down, improperly connected, or incorrectly configured can prevent communication even when IP addressing appears correct. Commands such as `show ip interface brief` were used to verify interface status and identify connectivity issues.

Proper cable management and device organization are also essential factors during physical installations. Organized cabling simplifies troubleshooting, reduces configuration mistakes, improves scalability, and supports long-term maintenance. In addition, network devices must have stable power connections and functioning hardware components before logical configurations can operate successfully.

During the lab, I configured network devices using the Cisco IOS command-line interface.

Key configuration tasks included:

- Accessing privileged EXEC mode using the command `enable`
- Entering global configuration mode with `configure terminal` command
- Assigning IP addresses to interfaces with `ip interface`
- Enabling interfaces using the `no shutdown` command
- Saving configurations using `copy running-config startup-config`
- Verifying interface status and connectivity

This lab required recalling commands and configuration sequences learned throughout previous networking exercises, reinforcing operational familiarity and long-term retention.

Troubleshooting was one of the most valuable parts of the activity for me, especially because it causes a bit of anxiety for me. Several common connectivity and physical interface issues were identified and corrected during the configuration process.

Common troubleshooting areas included:

- Incorrect cable placement
- Interfaces in an administratively down state
- Incorrect IP addressing
- Missing default gateways
- Improper interface configuration
- Devices connected to incorrect ports

To troubleshoot connectivity issues, I followed a structured process:

1. Verified physical cable connections
2. Checked interface status using `show ip interface brief`
3. Confirmed interfaces were enabled
4. Verified IP addressing accuracy
5. Used ping testing to validate communication
6. Reviewed device configurations for errors

When connectivity initially failed, I systematically reviewed interface states, verified addressing information, and corrected configuration mistakes until successful communication was established. This troubleshooting process reinforced the importance of validating Layer 1 and Layer 2 connectivity before assuming higher-level configuration problems exist.

The lab successfully demonstrated basic network connectivity between configured devices. Interfaces were enabled correctly, IP addressing was assigned properly, and end-to-end communication was verified through successful ping testing.

This activity strengthened my understanding of how:

- Physical connectivity
- Interface configuration
- Logical addressing
- Cabling standards
- Troubleshooting methodology

all work together to establish reliable network communication.

This lab was the most impactful activity within the competency because it required independently applying foundational networking knowledge rather than simply following instructions. It improved my confidence working within the Cisco CLI environment, strengthened my troubleshooting abilities, and reinforced the importance of structured problem-solving during network implementation and support.

Supporting evidence for this assessment includes:

- Packet Tracer topology screenshots
- Configuration screenshots
- CLI command outputs
- Connectivity verification results
- Walkthrough video documentation

I also appreciated the requirement to present recorded walkthrough material because it encouraged repetition and reinforced the learning process. Preparing for the recording required me to repeat the lab activities multiple times, practice the configuration steps, and recall commands from memory rather than relying entirely on written instructions. This process strengthened my understanding of the concepts, improved command familiarity within the Cisco CLI environment, and increased my confidence troubleshooting and configuring network devices independently.

Primary Basic Network Connectivity assessment walkthrough (video):

<https://youtu.be/L36QagIXgeI?si=I7ASUJBfag6RFuq8>

Supporting physical layer connectivity walkthrough (video):

https://youtu.be/9D_paJi_k20?si=tPz9SXDn5ba-FYS4

<https://herbal-rancher.github.io/portfolio/>